



Part—I

Programming

Using Glade and Perl

This article is the first in a series that covers the use of Glade, the GTK User Interface designer tool, to craft GTK UIs programmed in Perl.

Glade is a tool for designing GTK/ GTK+ GUIs, and is used quite a bit by developers who work with the GNOME desktop environment. It has been around for a long time. This tutorial uses Glade 3.6.7, the latest stable version at the time of writing, on Fedora 13, as the reference platform. However, it should work with all recent Linux distributions.

Check that you have Glade version 3 installed, with the following command:

```
$ glade-3 --version  
glade3 3.6.7
```

If you see something like the above output (it doesn't matter if the minor version number is different, ensure that you have at least 3.6) then you have Glade 3 installed. If you get an error message, then either you don't have Glade installed, or have an earlier version. Install Glade 3 by running the following command:

```
$ sudo yum install glade3
```

Getting started

Our tutorial project will use Glade to design an application with two screens—a 'user authentication' screen and an 'employee information viewer'. In this article, we will simulate some of the functionality of a real project; we won't actually use a database, since we are only demonstrating how to design the UI, and will write the bare minimum code to get it 'working'.

Once you fire up Glade, select *GtkBuilder* for the project file format, and choose to have object names unique within the project. For the GTK+ Catalog option, choose version 2.16.

GTK is different from the older (pre .NET) Windows 'fixed-position' layout model for GUIs, where you placed controls (widgets) on a form using absolute positioning and sizing, and then had to write extra code to reposition or resize the controls – in response to events like the user resizing the window, or changing screen resolution. In GTK, controls generally try to occupy as much space as they can in their parent container. If the container is resized, the controls inside it automatically